Final Business Report

Credit Risk Prediction Using Machine Learning

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1. Executive Summary

Financial institutions face significant risk from loan defaults, which can lead to financial losses and operational inefficiencies. Early identification of high-risk borrowers is critical for improving lending decisions and minimizing credit losses.

This project develops a **machine learning-based credit risk prediction model** that analyzes customer financial behavior and repayment history to predict the likelihood of loan default. Multiple machine learning models were trained and evaluated, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and AdaBoost. Among these models, **Random Forest demonstrated the best predictive performance with an ROC-AUC score of 0.833**, indicating strong ability to distinguish between default and non-default borrowers.

The results show that financial behavior indicators such as **payment history, customer age, repayment span, and debt recovery metrics** play a significant role in predicting default risk. The model can support lenders in making more informed credit approval decisions and reducing financial risk.

2. Business Problem

Credit lending institutions must evaluate whether a borrower is likely to repay a loan. Incorrect credit decisions can lead to:

- Increased loan defaults
- Financial losses
- Higher operational costs
- Reduced portfolio quality

Traditional credit assessment methods rely heavily on rule-based systems and manual evaluation, which may fail to capture complex behavioral patterns.

The objective of this project is to build a **machine learning model capable of predicting credit default risk based on historical financial behavior data**, enabling lenders to make more accurate and data-driven decisions.

3. Data Understanding

The dataset used in this project contains **customer financial behavior indicators related to credit usage and repayment patterns**.

Key variables in the dataset include:

- Repayment history
- Credit account status
- Debt recovery indicators
- Merchant category activity
- Payment span behavior

- Account balances and transaction patterns

These variables provide insights into borrower financial habits and repayment behavior, which are essential for identifying potential default risk.

4. Data Preparation

Data preprocessing steps were performed to ensure the dataset was suitable for machine learning modeling.

Key preprocessing steps included:

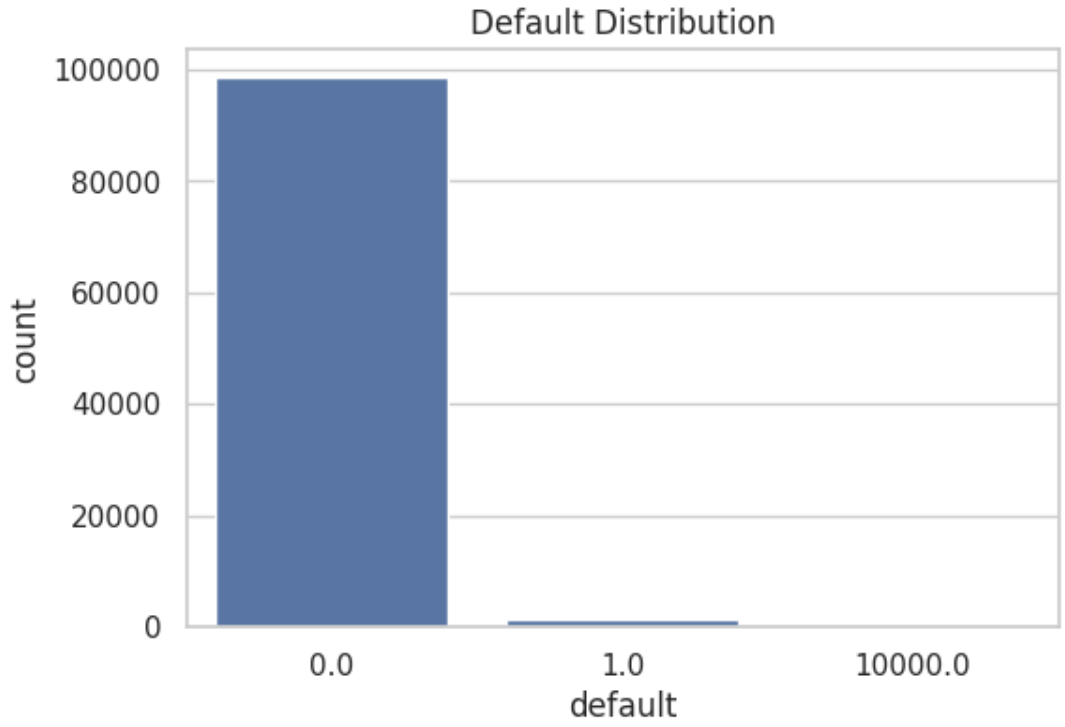
- Handling missing values
- Encoding categorical variables
- Scaling numerical features using **StandardScaler**
- Splitting the dataset into **training and testing sets**
- Ensuring proper distribution of the target variable

These steps helped improve model performance and ensured reliable predictions.

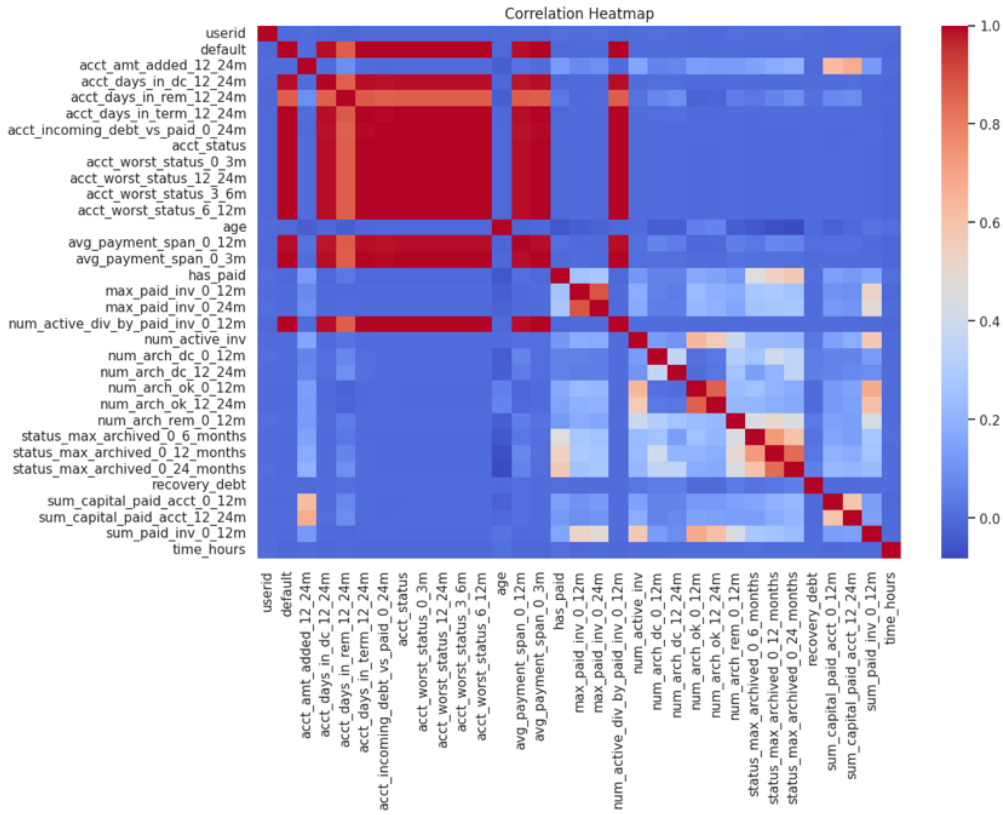
5. Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to understand relationships between variables and identify patterns associated with credit default.

Distribution of default vs non-default borrowers



Correlation heatmap of financial behavior features



Key EDA findings include:

- Certain repayment behavior variables show strong correlation with default risk.
- Customers with irregular repayment patterns are more likely to default.
- Payment span and debt recovery metrics influence credit risk significantly.

Visualization techniques such as **distribution plots and correlation heatmaps** were used to analyze these relationships.

6. Machine Learning Models

The following machine learning algorithms were implemented and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- AdaBoost

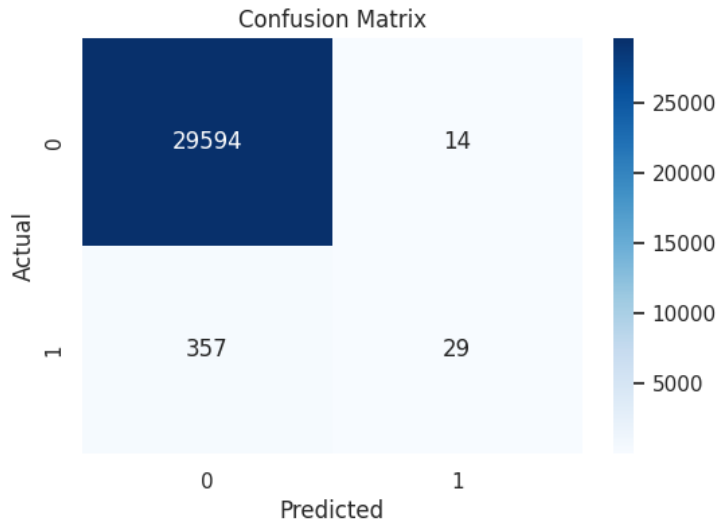
Each model was trained on the prepared dataset and evaluated using performance metrics such as **ROC-AUC and confusion matrix analysis**.

Among these models, **Random Forest achieved the best predictive performance**.

7. Model Evaluation

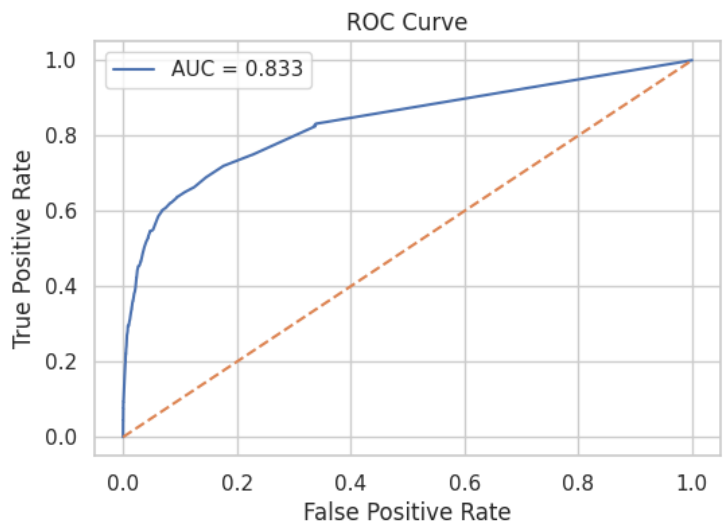
The performance of the Random Forest model was evaluated using a confusion matrix and ROC-AUC score.

Confusion matrix showing model prediction results



Model Performance Metrics

Metric	Value
ROC-AUC Score	0.833
True Negatives	29,594
False Positives	14
False Negatives	357
True Positives	29

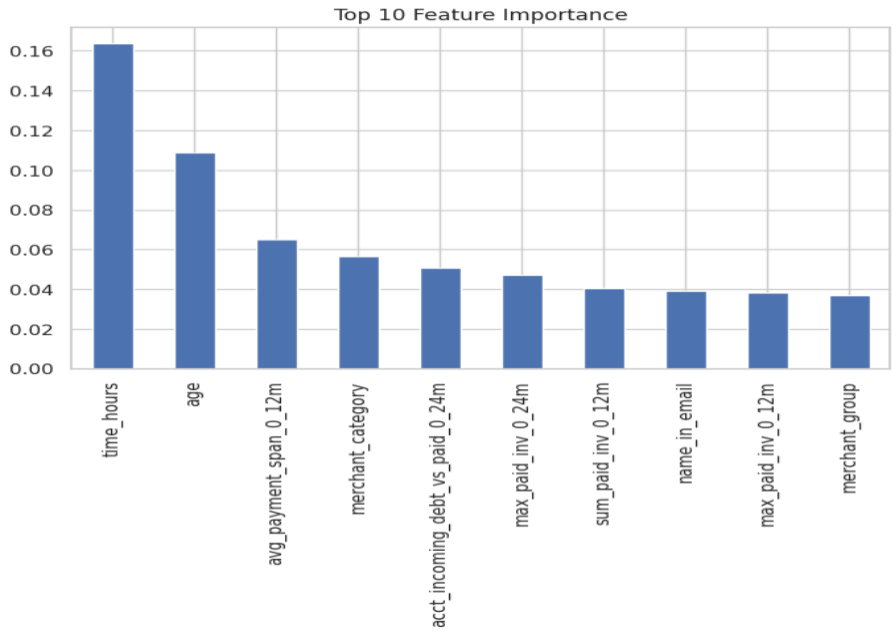


ROC curve showing model performance (AUC = 0.833)

The model demonstrates a **strong ability to correctly identify non-default borrowers while maintaining predictive capability for detecting potential defaults**.

8. Key Predictive Features

Feature importance ranking from Random Forest model



Feature importance analysis identifies the variables that most strongly influence credit default prediction.

Feature importance analysis revealed the following variables as the most influential predictors of credit default:

- Time since last financial activity
- Customer age
- Average payment span
- Merchant category behavior
- Incoming debt vs paid ratio
- Invoice payment history

These variables capture borrower financial discipline, spending behavior, and repayment consistency.

9. Business Insights

The analysis reveals several important insights:

1. **Repayment behavior is the strongest predictor of default risk.**
2. **Customers with irregular payment spans or delayed repayments exhibit higher default probability.**
3. **Financial activity patterns can provide early warning signals of potential default.**

4. **Machine learning models significantly improve credit risk detection compared to traditional rule-based systems.**

10. Business Recommendations

Based on the model results, financial institutions can implement the following strategies:

1. **Risk-Based Lending**

Use the prediction model to classify borrowers into low-risk and high-risk categories.

2. **Improved Loan Approval Decisions**

Integrate machine learning predictions into credit approval workflows.

3. **Early Risk Detection**

Identify high-risk customers early and implement intervention strategies.

4. **Dynamic Credit Scoring**

Continuously update credit risk scores using behavioral data.

11. Conclusion

This project demonstrates the effectiveness of **machine learning techniques in predicting credit default risk using financial behavior data**.

Among the evaluated models, **Random Forest achieved the highest predictive performance with an ROC-AUC score of 0.833**.

The model successfully identifies key behavioral indicators associated with borrower default risk and can be integrated into lending systems to support better credit decision-making.

By leveraging machine learning-driven insights, financial institutions can reduce credit losses, improve portfolio quality, and make more data-driven lending decisions.